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Working Prototype Overview

Video Demonstration

Working Prototype Video Demonstration 1: https://youtu.be/9d6GbXNmM6Y
Working Prototype Video Demonstration 2: https://youtu.be/H2iKPrMXN8s

Electronics Integration

The two videos provided above outline the integration of the electronics in this controller prototype comprehensively. Figures 1 and 2 visualize the controller showcased in the videos.

Figure 1
An image of the exterior of the controller prototype

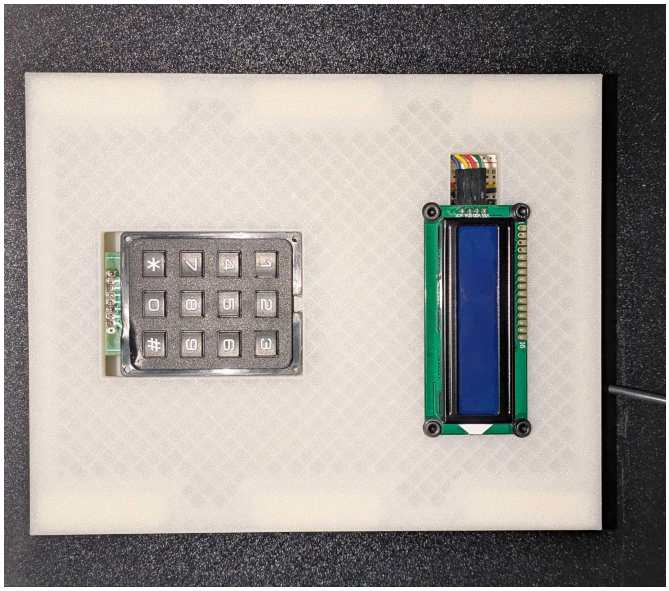
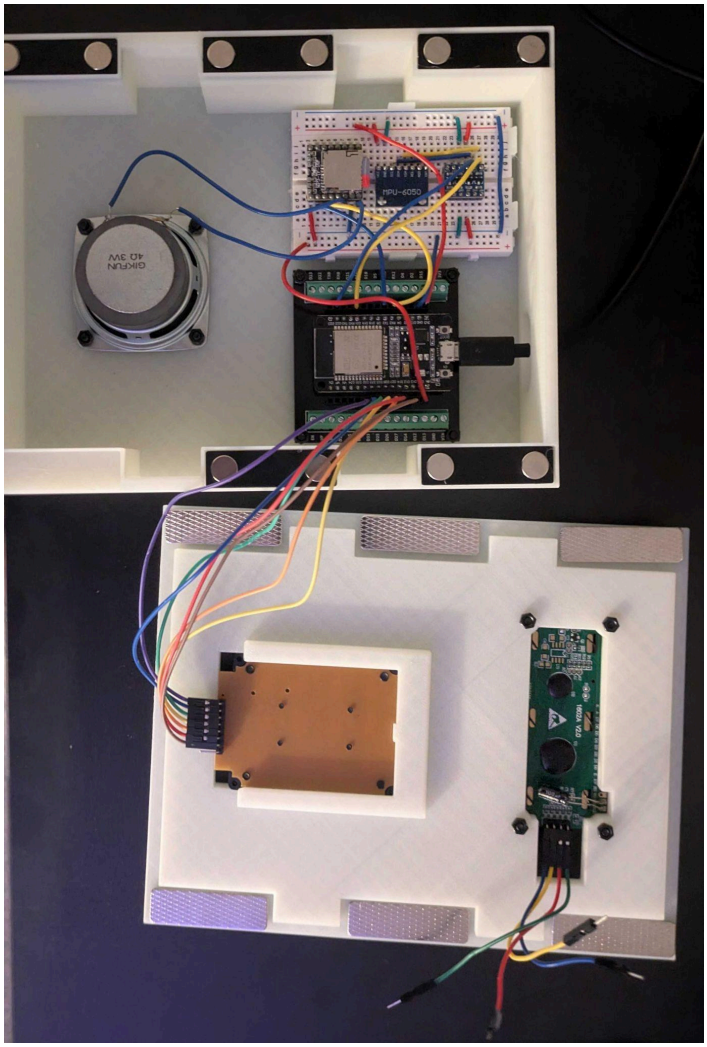


Figure 2

An image of the interior of the controller prototype

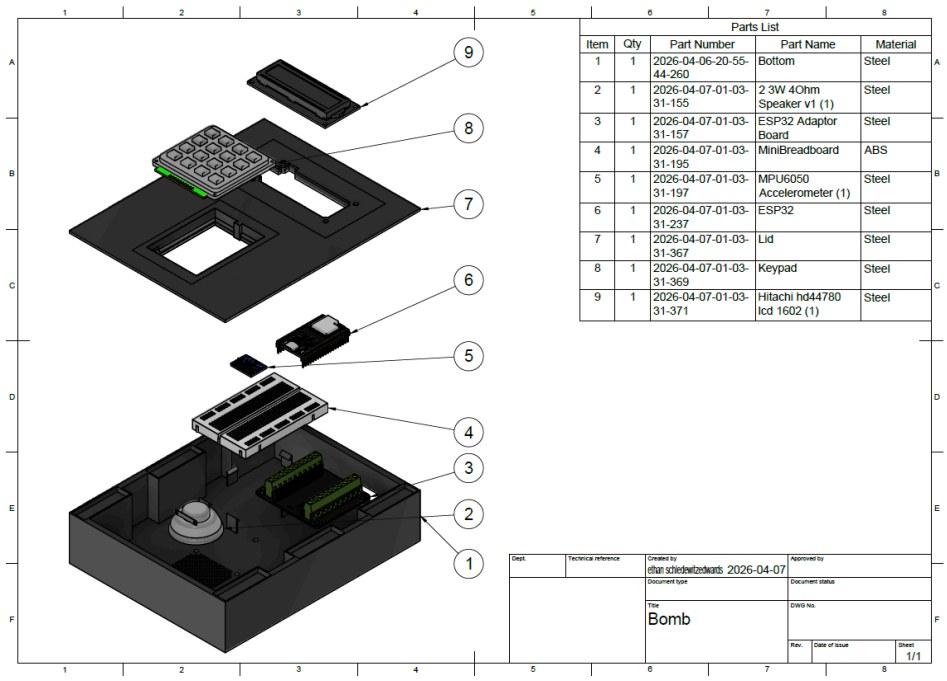


Bill of Materials

After the design of the Lid and Chassis of the controller was finalized, the Bill of Materials visualized in Figure 3 was created using Fusion 360's drawing tools.

Figure 3

The final Bill of Materials



Note. Some of the models used to represent the electronic components are incorrect and were used as approximations for the exploded view and bill of materials. The incorrect components are the matrix keypad and LCD screen.

References

[LCD]
https://grabcad.com/library/hitachi-hd44780-lcd-1602-1

[ESP32]
https://grabcad.com/library/esp32-devkit-32u-with-external-antenna-and-3-types-of-adapter-boards-1

[Breadboard]
https://grabcad.com/library/minibreadboard-2

[Accelerometer]
https://grabcad.com/library/mpu6050-accelerometer-module-2

[Speaker]
https://grabcad.com/library/3w-4ohm-speaker-1

[Keypad]
https://grabcad.com/library/4x4-button-keypad-matrix